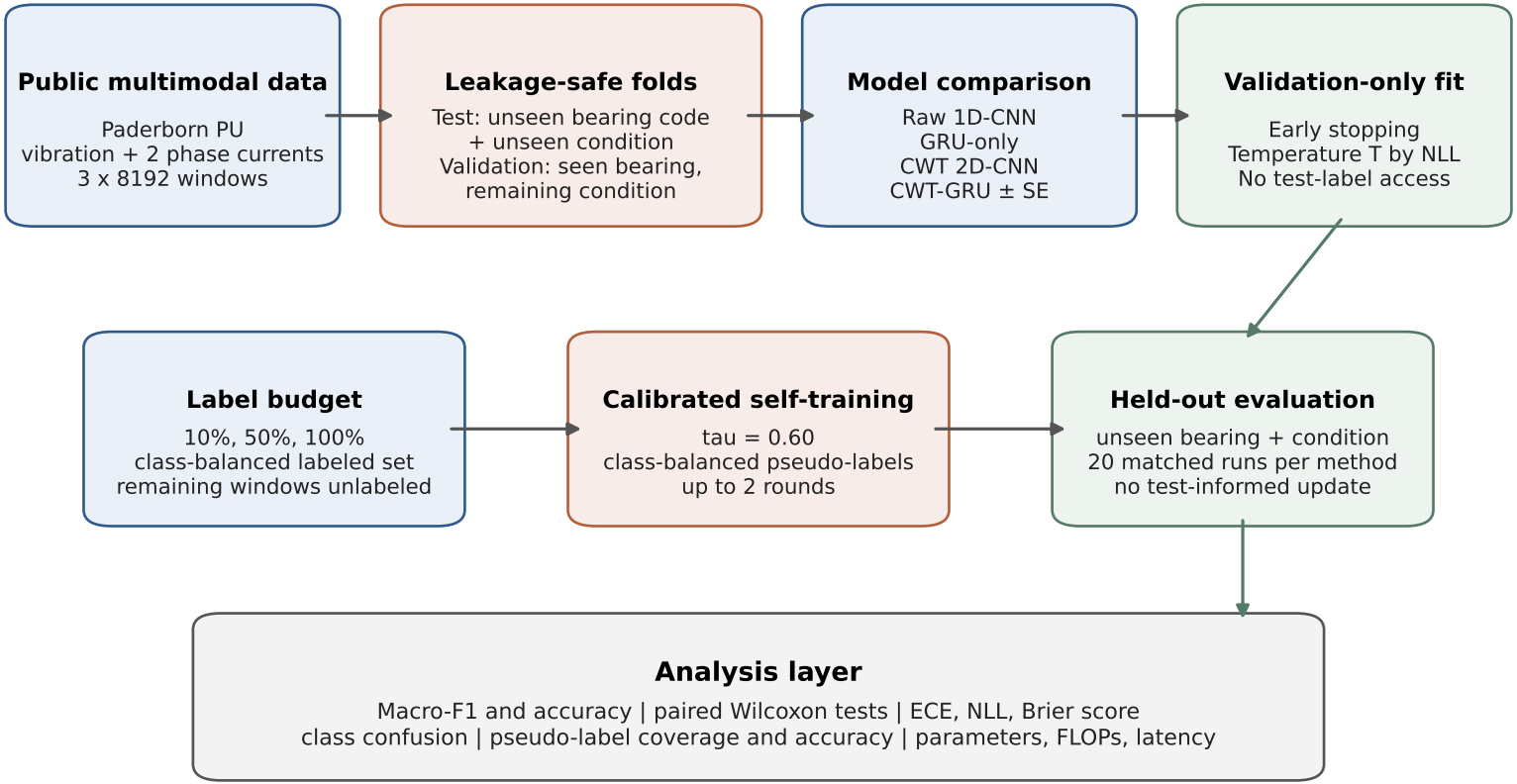
Public multimodal data
Paderborn PU
vibration + 2 phase currents
3 x 8192 windows
Leakage-safe folds
Test: unseen bearing code
+ unseen condition
Validation: seen bearing,
remaining condition
Model comparison
Raw 1D-CNN
GRU-only
CWT 2D-CNN
CWT-GRU ± SE
Validation-only fit
Early stopping
Temperature T by NLL
No test-label access
Label budget
10%, 50%, 100%
class-balanced labeled set
remaining windows unlabeled
Calibrated self-training
tau = 0.60
class-balanced pseudo-labels
up to 2 rounds
Held-out evaluation
unseen bearing + condition
20 matched runs per method
no test-informed update
Analysis layer
Macro-F1 and accuracy | paired Wilcoxon tests | ECE, NLL, Brier score
class confusion | pseudo-label coverage and accuracy | parameters, FLOPs, latency